A Survey on Driver Behavior Analysis From In-Vehicle Cameras

Jiyang Wang, Weiheng Chai, Archana Venkatachalapathy, Kai Liang Tan, Arya Haghighat, Senem Velipasalar, *Senior Member, IEEE*, Yaw Adu-Gyamfi, and Anuj Sharma

***Abstract*— Distracted or drowsy driving is unsafe driving behavior responsible for thousands of crashes every year. Studying driver behavior has challenges associated with observing drivers in their natural environment. The naturalistic driving study (NDS) has become the most sought-after approach, since it eliminates the bias of a controlled setup, allowing researchers to understand drivers' behavior in real-world scenarios. Video recordings collected in NDS research are incredibly insightful in identifying driver errors. Computer vision techniques have been used to autonomously analyze video data and classify drivers' behavior. While computer vision scientists focus on image analytics, NDS researchers are interested in the factors impacting driver behavior. This survey paper makes a concerted effort to serve both communities by comprehensively reviewing studies, describing their data collection, computer vision techniques implemented, and performance in classifying driver behavior. The scope is limited to studies employing at least one camera observing the driver inside a vehicle. Based on their objective, papers have been classified as detecting low-level (e.g. head orientation) or high-level (e.g. distraction detection) driver information. Papers have been further classified based on the datasets they employ. In addition to twelve public datasets, many private datasets have also been identified, and their data collection design is discussed to highlight any impact on model performance. Across each task, algorithms employed and their performance are discussed to establish a baseline. A comparison of different frameworks for NDS video data analytics throws light on the existing gaps in the state-of-the-art that can be addressed by future computer vision research.**

***Index Terms*— Driver behavior analysis, gaze, face detection, head orientation, drowsiness, driver distraction, lane change, survey.**

I. Introduction

THE driver, vehicle, and road form a close-knit, interactive system, within which safety and mobility are both of utmost importance and significantly affected by driver behavior [1]. Traffic crashes are the biggest cause of concern in a road environment. In 2019, the United States Department of Transportation's National Highway Traffic Safety Administration (NHTSA) reported 36,096 fatalities and 2.74 million injuries due to motor vehicle crashes [2]. Although the fatalities have decreased since 2016, the overall number of crashes has increased, and efforts are underway to achieve the ultimate goal of "Zero Fatalities".

Before 2002, traffic safety research relied heavily on police reports of traffic fatalities and injuries, and on healthcare and insurance databases. One such study, the 1977 Indiana tri-level study [3], reported that the number of accidents due to driver errors was higher than the number due to environmental and vehicle factors. Indeed, human error was identified in this study as the definite or plausible cause in more than 90% of the accidents observed. This study also identified specific behavioral characteristics, such as reduced alertness (23%), inattention (17%), excessive speed (10%), improper evasive action (13%), and internal distraction (9%), as the leading causes of accidents.

Although there is strong evidence that human error is a plausible cause in most crashes, only a handful of studies have observed the same through in-vehicle driver behavior in their natural environment. This was due to technological constraints before the 2000s. However, with the increasing availability of low-cost sensor and storage devices, naturalistic driving studies (NDS) have become the most sought–after approach to provide insight into drivers' everyday behavior. NDS vehicles are equipped with sensors –allowing for collecting vehicle and driving parameters such as speed, braking, or acceleration– and cameras for recording all events inside and outside the vehicle. The advantage of NDS research lies in the broader perspective it offers towards understanding normal traffic behavior in natural settings, and its holistic insights into the actions that lead to a crash or a near-miss. It eliminates the perception bias induced in simulator studies, which are performed under a controlled setup, and limited in their potential to replicate the uncertainties of traffic environment. In addition, when subjects are aware of or think that they are participating in an experiment, this may affect their driving behavior [4]. The camera positioned inside vehicles in an NDS approach provides valuable information about drivers' alertness, drowsiness/fatigue, distractions, and other behavioral cues before they perform a driving operation or maneuver.

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Jiyang Wang, Weiheng Chai, and Senem Velipasalar are with the Department of Electrical Engineering and Computer Science, Syracuse University, Syracuse, NY 13244 USA (e-mail: jwang127@syr.edu; wchai01@syr.edu; svelipas@syr.edu).

Archana Venkatachalapathy, Arya Haghighat, and Anuj Sharma are with the Department of Civil, Construction and Environmental Engineering, Iowa State University, Ames, IA 50011 USA (e-mail: archanav@iastate.edu; aryahagh@iastate.edu; anujs@iastate.edu).

Kai Liang Tan is with the Department of Mechanical Engineering, Iowa State University, Ames, IA 50011 USA (e-mail: kailiang@iastate.edu).

Yaw Adu-Gyamfi is with the Department of Civil and Environmental Engineering, University of Missouri, Columbia, MO 65211 USA (e-mail: adugyamfiy@missouri.edu).

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The 100-Car Naturalistic driving study [5] was the first large-scale study to observe drivers in their naturalistic environment. Following this, Virginia Tech Transportation Institute (VTTI) initiated their second and largest ever naturalistic driving study, the Second Strategic Highway Research Program Naturalistic Driving Study (SHRP 2 NDS) [6]. This study collected 2 Petabytes of continuous naturalistic driving data over three years from more than 3300 instrumented vehicles and 3500 participants across six different locations in the U.S. Dingus *et al.* [7] have analyzed SHRP 2 NDS data and reported that driver-related factors such as error, impairment, fatigue and distraction are responsible for 90% of crashes. Additionally, their paper noted that distractions due to the use of handheld devices increased driver risk. Subsequent analysis of the SHRP 2 NDS data has revealed several critical findings related to crash or near-crash events [8]–[10], distracted drivers [11]–[13], drowsiness [14]–[16], and fatigue [17].

Inspired by the SHRP 2 NDS, similar studies have been initiated worldwide, with China, Australia, and Canada expected to produce data at the same scale. The success of NDS research in increasing our understanding of human behavior has led to its expansion and use for studying the driving behavior of vulnerable road users and autonomous vehicle features. One of the major challenges in any large-scale NDS research is the effort required to reduce the massive datasets produced. NDS research needs robust infrastructure and arduous manual labor to extract meaningful information. The most time-consuming task of data reduction is extracting useful elements by annotating video datasets. Annotation of driver behavior of interest, such as a driver being distracted or performing a specific activity, is essential to understand the impact of drivers' behavior on driving risks as well as their response to an unexpected event.

Recent developments in computer vision and machine learning algorithms hold the key to accelerating NDS video data annotation with minimal human effort. These algorithms can autonomously analyze camera data, and extract essential information about drivers' behavior and traffic environment. Initially, algorithms that relied on handcrafted features were used for such detection tasks. In these earlier methods, Histogram of Oriented Gradients (HOG) [18], AdaBoost cascade detectors [19], and localized gradient orientation [20] were used for feature extraction, and Support Vector Machines, K-nearest neighbors (KNNs) [18], and Gaussian mixture models [21] were used for classification. However, the use of handcrafted features by such methods has resulted in their lack of transferability and lower accuracy, and hindered their adoption in processing naturalistic driving data, as evident in the minimal application of such techniques for NDS research. Deep learning techniques, such as Convolutional Neural Networks (CNNs), have been touted to address the aforementioned deficiencies. CNN-based models can extract features from any input data, and be trained to perform classification and detection with high accuracy [22]. CNNs also remove the need for designing hand-crafted features, which usually do not generalize well. With such capabilities, the deep model-driven data reduction can potentially revolutionize the automatic extraction and classification of driver behavior.

Traditionally, NDS researchers and computer vision scientists have been two different communities, though they have often worked in parallel to understand driver behavior. While computer vision scientists focus on detecting driver behavior, NDS researchers study the factors impacting the same. Previous review studies have addressed a subset of problems under traffic safety research. For example, Melnicuk *et al.* [23] have provided a review of driver state monitoring (DSM), wherein the driver state covers cognitive distraction, mental workload, mental fatigue and emotions, and also proposed a system for DSM. Authors have reviewed papers, which employ input data, such as heart activity, blood pressure, images, or their different combinations, without going into the details of different datasets and performance comparison. Loce *et al.* [24] have presented a review of computer vision techniques in three domains of transportation: safety, efficiency, security and law enforcement. However, the authors focused on built-in smart vehicle applications and not the intrinsic driver behavior associated with each domain. Another direction to such studies has lain in identifying driver behavior by detecting low-level (e.g., eyes, head orientation, mouth, hands) [25], [26] or high level (e.g., distraction or drowsiness) [27]–[29] information about drivers. Such studies have compared the implementation and performance of different computer vision algorithms.

Although the reviewed studies have dealt with similar objectives, they have differed in the type of NDS dataset they handled, variables studied, and models deployed. There has been little to no review conducted across these studies of their datasets and data collection efforts, affecting the transferability of the analyses they have presented. Dividing these studies according to data collection techniques, such as the number of cameras used, camera position, resolution and sample size, can further enhance the objective of extracting from such studies detailed information about driver behavior. Besides detection, computer vision algorithms can also be used for other steps of NDS research. Thus, there has been a gap in understanding the ability of the computer vision techniques employed to study driver behavior, and a holistic effort is needed to identify the state-of-the-art practices for using computer vision algorithms to also automatically extract relevant driver behavior information from the naturalistic driving data. The main objectives of this survey paper are therefore three-fold:

1) Establish a baseline accuracy in using deep models to extract an event of interest in NDS - This will help researchers choose from a range of techniques that have been demonstrated to achieve acceptable accuracy for autonomously extracting relevant driver behavior information.
2) Understand the impact of sensor placement on performance accuracy - This will help new NDS to choose the most appropriate placement of sensors for improved data collection.
3) Establish gaps in the current state-of-the-art that can be addressed by future computer vision research.

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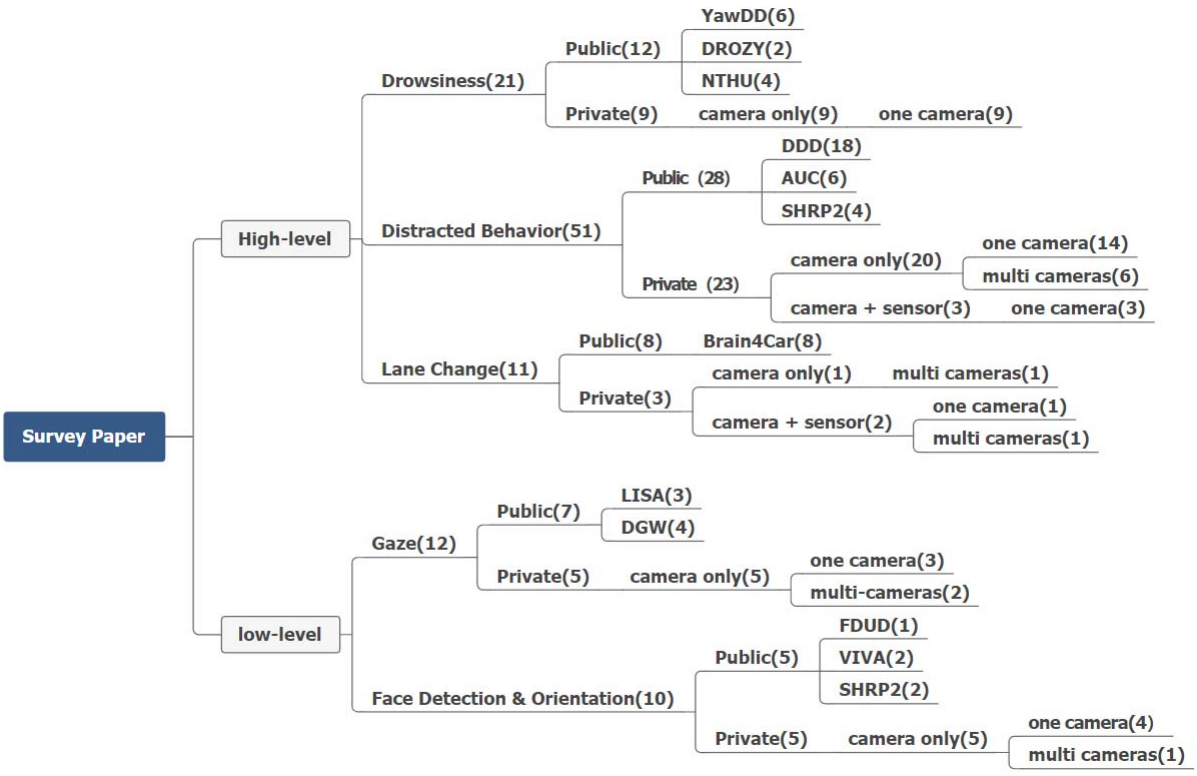


Fig. 1. Categorization of the reviewed papers. The number of papers covered for each leaf node is included in parentheses.

The scope of this paper is limited to studies using at least one in-vehicle camera facing the driver, and employing real cars and cameras. We will first dissect the driving behavior analysis in terms of the corresponding low-level and high-level analysis tasks. Under each task, a collection of NDS data and papers are reported in terms of their respective objective, techniques, and performance. The datasets have been broadly categorized as public versus private datasets. Studies using public datasets are compared in terms of their performance, benchmarking, and computer vision techniques used in studying driver behavior. As a final step, a detailed discussion is provided, which includes a synthesis of insights into the findings obtained from this systematic review in regards to our three study objectives.

A. Categorization of the Reviewed Papers

Across the literature, there has been a large variety of approaches using different datasets collected via different combinations of cameras and sensors, and focusing on various driver tasks. While categorizing these papers, it is important to pay attention to the different types of datasets employed so that readers can compare the performances of methods using the same dataset.

In this survey paper, the reviewed papers are categorized according to the chart presented in Fig. 1. First, papers are classified into two categories, namely those focusing on low-level tasks versus high-level tasks, which are described in Sections II and III, respectively. This classification is based on the semantic complexity of each analysis and detection task. More specifically, the low-level task category covers papers focusing on (i) gaze detection (Sec. II-A) and (ii) face and head orientation detection (Sec. II-B). On the other hand, the high-level task category covers papers focusing on the (i) detection of driver drowsiness (Sec. III-A), (ii) detection of distracted driver behavior (Sec. III-B), and (iii) prediction/detection of lane changes (Sec. III-C). For example, drowsiness can be detected by analyzing physical and physiological changes through visual and non-visual features. Distracted behavior is defined as a driver interacting with objects other than the steering wheel. High-level tasks are built from several low-level tasks, e.g. eye opening is an important feature for drowsiness detection. The classes in the high-level task category cover the important scenarios for driver analysis.

For each of the aforementioned categories, the papers reviewed are further categorized based on the dataset(s) that they use. The datasets may be public/benchmark datasets or private datasets captured by the authors of particular papers. We provide such categorization so that it is possible to compare the performances of papers focusing on the same task and using the same dataset. For instance, the papers focusing on drowsiness detection and using public datasets are described in Sec. III-A.1, and the papers focusing on drowsiness detection and using private datasets are described in Sec. III-A.2. For the public datasets, the majority of the papers reviewed have used the StateFarm Distracted Driver Dataset (DDD) [30], AUC (American University in Cairo) distracted driver dataset [31], Brain4Cars [32], SHRP2 dataset [33], FDU Drivers dataset [34], Vision for Intelligent Vehicles & Applications (VIVA) dataset [35], ZJU-dataset [36], Southeast

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TABLE I
COMPARISON OF THE PUBLIC DATASETS USED FOR DRIVER GAZE ESTIMATION

	$LISA\ V_0$ [47]	$LISA\ V_1$ [48]	$LISA\ V_2$ [46]	DGW [56]
Year	2017	2018	2020	2020
No. of frames	47515	47515	336177	586 videos
No. of views	3	3	1	1
No. of participants	11	11	N/A	338
Male/Female	N/A	7/4	N/A	247/91
No. of classes	7	7	7	9
Camera Type	RGB	RGB	IR	RGB
Resolution	2704x1524	2704x1524	640x480	179x179

University driving-posture dataset (SEU) [37], YawDD [38], DROZY [39], and NTHU [40]. We describe each of these datasets below in detail. As seen in Fig. 1, the papers using private datasets are further categorized based on whether they use only cameras or a combination of cameras and other sensors (e.g., to identify GPS location or speed). Then, one final level of subdivision was performed based on whether a study employed a single camera or used multiple cameras. In the chart shown in Fig. 1, the number of papers considered in this review from each category is listed in parentheses. For instance, this survey paper covers 18 papers focusing on detecting distracted drive behavior and using the StateFarm Dataset (DDD).

II. LOW-LEVEL TASKS

This category covers papers focusing on low-level tasks, more specifically (i) gaze detection and (ii) face and head orientation detection. Learning to detect face, eye and head pose are important components, often considered as initial steps in the detection or classification of high-level driver information. Papers focusing on gaze detection and head orientation detection will be summarized in Sections II-A and II-B, respectively.

A. Driver Gaze Estimation

Driver's gaze is a critical piece of information for understanding the driver's state. It is often used for determining the driver's attention [41], readiness to take over control from partially automated vehicles [42] or simply to classify their zone of vision [43]–[55]. Among the papers reviewed, some used two public datasets, namely the LISA Gaze Dataset (with three different versions) [46]–[48] and Driver Gaze in the Wild dataset [43]–[45], [56], while others used privately collected datasets [49]–[52], [57].

1) Studies Using Public Datasets: Three versions of the publicly available LISA Gaze Dataset have been employed by several studies. This dataset became publicly available very recently and can be downloaded from GitHub [46]. The three different versions of the LISA Gaze Dataset, and the Driver Gaze in the Wild (DGW) dataset are summarized and compared in Table I. The rest of this section describes each of these data sets in more detail.

- **LISA Gaze Dataset** V_0 was presented by Vora *et al.* [47] for driver gaze estimation. Its data were captured by using two cameras facing the driver, as well as one camera pointing outside. The cameras facing the driver captured the driver's face from different perspectives. One was mounted near the rear-view mirror while the other was mounted near the A-pillar on the side window. In [47], only images from the camera mounted near the rear-view mirror were used for the experiments. The other views were provided to a human expert for labeling the ground truth gaze zone. The different forms of driver gaze were annotated using seven classes, which include forward, right, left, center stack, rear-view mirror, speedometer gaze, and eyes closed. A Convolutional Neural Network (CNN)-based method was used for gaze estimation. The data were preprocessed in three different ways:
  1) The driver's face was detected and used as an input
  2) Some context was added to the driver's face by extending the face bounding box in all directions
  3) Only the top half of the face was used as an input

  Vora *et al.* [47] have used the LISA Gaze Dataset to fine-tune AlexNet [58] and VGG16 [59], both of which had been pretrained on ImageNet [60]. The best-performing model, using only the top half of the face as input, was VGG16, which achieved an accuracy of 93.36% and showed good generalization capability.
- **LISA Gaze Dataset** V_1 was also introduced by Vora *et al.* [48]. Compared to V_0, it provides additional information about weather, time of drive, drivers' age and gender. The images were preprocessed in four different ways:
  1) The driver's face was detected and used as an input
  2) Some context was added to the driver's face by extending the face bounding box in all directions
  3) Only the top half of the face was used as an input.
  4) The driver's surroundings, referred to as the face-embedded field of view (FoV), was used as an input.

  In [48], only one view from this dataset was used to fine-tune the AlexNet, VGG16, ResNet50 [61] and SqueezeNet [62], all of which had been pretrained on ImageNet. The best-performing model was found to be SqueezeNet, using only the top half of the face as input, and achieving an accuracy of 95.18%.
- **LISA Gaze Dataset** V_2 was introduced by Rangesh *et al.* [46]. Data were captured by an Intel RealSense infrared IR) camera mounted next to the rear-view mirror. The classes in V_2 of the LISA Gaze Dataset are the same as in V_0 and V_1. The labels in V_2 also have information about the time of drives (daytime or nighttime) and about whether drivers were wearing glasses or not. In real-world naturalistic driving scenarios, drivers may wear glasses and/or there might not be sufficient light in the cabin, especially at night. Rangesh *et al.* [46] have proposed a gaze preserving CycleGAN [63] to address the eyeglass issue in driver gaze estimation and also used an infrared camera to overcome the lighting issue. They preprocessed the data using the gaze preserving CycleGAN to remove the glasses and then trained SqueezeNet. Their reported accuracy with the gaze-preserving CycleGAN was

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TABLE II
COMPARISON OF THE PAPERS USING THE LISA GAZE DATASET FOR DRIVER GAZE ESTIMATION

Paper	Year	Dataset	Method	Accuracy
Vora et al. [47]	2017	LISA V_0	VGG16	93.36%
Vora et al. [48]	2018	LISA V_1	SqueezeNet	95.18%
Rangesh et al. [46]	2020	LISA V_2	Cyclegan + SqueezeNet	72.84%

TABLE III
COMPARISON OF THE PAPERS USING DGW DATASET FOR DRIVER GAZE ESTIMATION

Paper	Year	Method	Accuracy
Stappen et al. [45]	2020	VGG16	93.36%
Lyu et al. [44]	2020	SqueezeNet	95.18%
Yu et al. [43]	2020	Cyclegan + SqueezeNet	72.84%

80.94%, a higher accuracy compared to using their input data directly to train SqueezeNet, which provided an accuracy of 72.84%. The comparison of works using the LISA dataset is provided in Table II.

- **The Driver Gaze in the Wild (DGW) Dataset** has been presented by Ghosh *et al.* [56] for driver gaze zone estimation, and was labelled autonomously using the speech-to-text conversion technique. This dataset includes 586 videos of 338 different subjects varying between 18 to 63 years of age. The videos were recorded with a single Microsoft Lifecam RGB camera equipped with a microphone. The subjects were directed to look at nine different zones inside the vehicle such as the back mirror, side mirrors, radio, speedometer and windshield. The dataset contains videos across different illumination settings (i.e., 55.2% in daylight and 44.8% in evening light) and weather conditions. The average size of the driver's face in the DGW image is 179×179 pixels. The dataset also accounts for subjects with glasses, recording both scenarios with and without glasses. Each time the subject was asked to gaze in the direction of a particular zone, the audio recorded includes the number designated for the zone. This audio file was extracted and processed to identify the zone numbers that were then used to automatically label the dataset for driver gaze.

  At the Eighth Emotion Recognition in the Wild (EmotiW) 2020 challenge, the DGW dataset was used for building several driver gaze prediction models. For example, X-AWARE is a contextual attention mechanism [45] that combines the visual environment with the driver's face to model different features for gaze prediction. The objective was to identify facial features as well as important environmental context features towards modeling gaze prediction. The X-AWARE deep learning architecture had four main parts: two that used InceptionResNetV2 [53] (for extracting features from the full image and face image), Go-CaRD (Generic, optical Car Part Recognition and Detection) features extracted from the full image, and some manually crafted features. In this model, the outputs from the two InceptionResNetV2 models are fused. In parallel, the extracted features are enhanced and combined. Both streams are fine-tuned using categorical cross-entropy losses and combined to predict the final output. This model outperformed the baseline [56] by 15.03% accuracy on the validation set and reported a further increase by 8.72% on the test set.

  Another well-performing model that has been proposed by Lyu *et al.* [44] is a CNN-based multimodel fusion gaze zone estimation system. The architecture involves taking an image and using HOG and a linear Support Vector Machine (SVM) for facial landmark detection. Facial alignment in this model is carried out using affine transformation, perspective transformation and Delaunay triangulation transformation to normalize the face information. The InceptionV3-based classification network and a classifier network called PointNet are combined to predict the final output. On the DGW dataset, the proposed fusion model achieved an accuracy of 81.51%.

  Yu *et al.* [43] have proposed an effective multi-model fusion method for driver gaze estimation, which can overcome challenges due to illumination, pose and camera placement, and provided the best performance in the EmotiW 2020 challenge with a prediction accuracy of 82.5288%. The proposed model framework consists of two parts, namely, learning the appearance-based and geometry-based features. For appearance-based feature extraction, ResNet50, ResNeSt-50 [54] and EfficientNet [55] are fused. Moreover, a landmark model and Delaunay model are used for geometry-based features. These models are weighted using grid search optimization, and the fusion predictions are used for the final classification. The comparison of works using the DGW dataset for driver gaze estimation is provided in Table III.

2) Studies Using Private Datasets: In this section on studies that have addressed driver gaze estimation using private datasets, papers have been further classified based on whether the study described used a single camera or multiple cameras for the collection of driver data.

a) Studies using only one camera: Vasli *et al.* [49] employed a camera mounted near the rear-view mirror for gaze estimation. Each frame was classified into one of six categories: forward, right mirror, left mirror, infotainment panel, rear-view mirror and speedometer. The number of total annotated frames in this dataset is 13191, and 5364 frames were used in the experiments, 80% as training data and the remaining for testing. The authors have proposed a system to estimate the driver's gaze from head and eye cues projected onto a multi-plane geometrical environment, and a system that fuses this geometric analysis with the data-driven learning method. The reported accuracy is 95.29%.

Vicente *et al.* [51] have proposed an inexpensive vision-based system to accurately detect Eyes Off the Road (EOR). In this system, images are captured by a monocular camera that is mounted on the steering wheel column. The total number of frames is 13500, and the resolution of images is 1280×720 pixels. Fourteen off-road points are labeled, namely the driver window, driver mirror, driver visor, passenger visor, passenger window, passenger mirror, glove box, navigation system, steering wheel, windshield top left,

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windshield top center, windshield top right, windshield right and windshield bottom right. Four on-road points, namely windshield left, windshield center, windshield bottom left and windshield bottom, are also labeled. The system has three main components: 1) facial feature tracking; 2) head pose and gaze estimation; and 3) 3-D geometric reasoning to detect EOR. The authors have reported for this vision-based system an accuracy above 90%.

Naqvi *et al.* [52] have proposed a model to detect the gaze region of a driver by using an NIR camera placed on the dashboard. The captured images have a resolution of 1600×1200 pixels. 17 spots (gaze zones) were designated for drivers to gaze at for the experiment, and each driver stared at each spot five times. There were 20 drivers, and three of them wore glasses. The researchers trained a CNN model three times to detect and observe the left eye, the right eye and the face of the driver to detect the gaze region. The result of these experiments was reported in terms of strictly correct estimation rate (SCER) and loosely correct estimation rate (LCER). The authors reported 92.8% SCER and 99.6% LCER on their dataset.

b) Studies using multiple cameras: Birrell and Fowkes [57] presented a study investigating interactions of drivers with in-vehicle information systems while driving. They collected video data from 15 participants by using four cameras. One camera recorded driver's face, while three cameras monitored forward traffic, rear traffic and smartphone activity. These videos were then analyzed to determine drivers' glance, such as to mirrors, road, Foot-LITE smart driving system etc. They reported that, on average drivers spend 4.3% of their time looking at the smart driving system, with no glance lasting longer than 2s. The distribution of different glances and their duration helped them conclude that in-vehicle smart driving systems/interfaces do not lead to visual distraction. Tawari *et al.* [50] have presented a distributed camera framework to estimate a driver's coarse gaze direction using both head and eye cues. They used two cameras facing the driver for capturing the driver's movements. One camera was mounted near the rear-view mirror while the other was mounted near the A-pillar on the side window. These two cameras captured the face of the driver in color video at 30 fps with 640×360 resolution. The dataset includes 2000 annotated frames with six classes labeled: front windshield, center console, left window rear-view mirror, center rear-view mirror, right window rear-view mirror and speedometer. The authors have also proposed gaze-surrogate features estimated from the eye region via eyelid and iris analysis. The reported accuracy of their proposed method is 94.9%.

Gaze estimation is an essential component in driver behavior classification. Tracking the driver's gaze is critical to understanding distraction or drowsiness while driving. The movement of the eyes, zone of vision and opening of the eyes have been some of the important low-level features studied. In the two public datasets discussed, seven to nine different driver gaze zones within the vehicles have been identified that provide insight into driver behavior. The DGW dataset is the largest and most recently developed dataset covering all the critical gaze zones. Nevertheless, both of these public datasets possess rich and diverse information accounting for varying illumination and environmental conditions as well as drivers with and without glasses. In comparison, the listed studies using private datasets have analyzed a greater amount of sample data and number of classes. However, other information such as population size, weather conditions and illumination settings has rarely been available in studies using private datasets.

TABLE IV
COMPARISON OF PUBLICLY AVAILABLE DATASETS FOR FACE AND HEAD ORIENTATION DETECTION

	SHRP2 [33]	VIVA [35]	FDUD [34]
Year	2015	2016	2020
# of images	91k	N/A	20K
# of views	5	1	1
# of participants	41	N/A	100
Camera Type	GRAY/IR	RGB	N/A
Resolution	356x240	various$\times$554	N/A (HD images)

Studies using both private and public datasets have commonly focused on classifying the zone of driver's gaze. Their pipeline for learning has included cropping the driver's face, detecting facial landmarks to extract features of the eyes, and classification. With public datasets, deep learning models have frequently been employed to classify driver gaze, and achieved a substantial accuracy ranging between 80-95%. Studies using private datasets, on the other hand, have explored both traditional methods, such as Support Vector Machines, as well as deep models, such as CNNs, to estimate driver gaze. Across the different numbers and positions of cameras used, models have reported high accuracy values of above 90%. Thus, among the private and public datasets considered, the driver gaze estimation task has been thoroughly studied, with specific features and techniques that have been clearly identified as valuable for further use in high-level task detection.

B. Face Detection and Head Orientation Estimation

The detection of a driver's face is the first step towards analyzing the driver's behavior. Every feature of a driver's face holds potential significance towards understanding the driver's attention and concentration. Head orientation can be particularly informative as a behavioral cue before drivers execute decisions on the road. As a result, the low-level tasks of driver face detection and head pose estimation are useful in driver assistance systems [64], and can be used in other driver behavior analysis tasks, such as driver gaze estimation [65], distraction detection or drowsiness detection. This section will summarize studies that have focused on driver face detection and head pose estimation, again classifying investigations based on the datasets that they have used.

1) Studies Using Public Datasets: This section describes three key publicly available datasets together with studies that have employed these datasets to detect drivers' faces and estimate their head orientation. These three public datasets are summarized and compared in Table IV, and studies analyzing them are summarized in Table V.

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TABLE V
COMPARISON OF PAPERS USING PUBLIC DATASETS FOR FACE DETECTION AND HEAD POSE ESTIMATION

Model	Year	Dataset	Method	Precision/ MAE/Accuracy
Wang et al. [66]	2015	SHRP2	Viola-Jones Face detector	Precision - 99-100%
Paone et al. [33]	2015	SHRP2	Government Off the shelf	Face detection Accuracy – 87.9% (night), 88.9% (transition), 83.8% (day)
Martin et al. [35]	2016	VIVA	Boosted Cascade ACF	Average Precision – 66.3
Yuen et al. [67]	2016	VIVA	Deep CNN	Head Pose Accuracy – 88%
Yang et al. [34]	2020	FDUD	CNN	MAE – 3.72 degrees

TABLE VI
GOVERNMENT-OFF-THE-SHELF (GOTS) HEAD POSE ESTIMATION PERFORMANCE IN TERMS OF MEAN ABSOLUTE ERROR (MAE) (IN DEGREES) PRESENTED IN PAONE *et al.* [33]

	MAE: Yaw	MAE: Pitch	MAE: Roll
Night	5.98	11.3	1.72
Transition	4.47	10.2	1.51
Day	5.80	9.43	1.94

- **SHRP2 dataset [33]** contains digital video, GPS position and heading, acceleration data, rotation rates, and ambient lighting information collected at rates varying from 1Hz to 15Hz. The SHRP2 video data was captured from four different views: forward view, face view (resolution of 356×240), lap and hand view, and rear view, recorded at 15 frames per second fps) and compressed into a single quad video. A subset of the SHRP2 dataset, called the clipped set, consists of 41 videos including the face view, 20 of them from static trials and 21 from dynamic trials. It also contains kinematic data, static and dynamic vehicle segments, trial data, participant's personal details, etc. Wang *et al.* [66] have proposed a methodology for detecting and estimating driver's head position and orientation from the SHRP2 videos. Their estimation pipeline included face detection, based on the algorithm introduced by Viola and Jones [68], spurious face elimination, and adaptive template-based head tracking. Their reported precision is 99-100%. Paone *et al.* [33] has achieved the same objective using commercial and open-source face pose estimation algorithms on the SHRP2 head pose validation dataset. They used Government-off-the-shelf (GOTS), GOTS tracked (G-T) and DLib [69] for face detection and head pose estimation. The highest face detection accuracy in the driving trial was reported by G-T, with an accuracy of 87.9% at night, 88.9% during the day-night transition period, and 83.8% during the day. For the head pose estimation task, it was the GOTS algorithm that reported the highest performance, as detailed in Table VI.
- **The VIVA Face Dataset [35]** is composed of selected images from daytime naturalistic driving data collected by the Laboratory for Intelligent and Safe Automobiles (LISA) at UCSD and naturalistic driving videos from YouTube. Images were selected from a total of 39 video sequences, based on harsh lighting conditions and facial occlusions, and were resized to the same height of 554 pixels by preserving the aspect ratio. Martin *et al.* [35] has described the Vision for Intelligent Vehicles and Applications (VIVA) platform, and presented various issues and challenges pertaining to naturalistic driving studies. The VIVA platform was designed with the intent to explore, compare and benchmark methods for the detection of the driver's face inside the cabin. The researchers tested publicly available face detection models with the dataset, and compared their results in terms of Precision-Recall curves. These models included Boosted Cascade with Aggregated Channel Features (ACF) [70], Boosted Cascade with Haar [71] and the Viola-Jones algorithm [68]. The highest average precision (AP) value of 66.3 was reported by the Boosted Cascade with ACF model.
  Advancing to deep learning models, Yuen *et al.* [67] designed a deep CNN-based face detection method with discrete head pose estimation, overcoming the limitations of harsh lighting conditions, occlusions and varying viewpoints. The performance of their proposed CNN was compared with other benchmark studies of the VIVA dataset using Precision and Recall curves. Their proposed system performed better than the baseline methods even when there is occlusion. Their proposed deep CNN could also classify head pose with an accuracy of 88%.
- **Fudan University Driver Dataset (FDUD)** was introduced by Yang *et al.* [34] for use in the development of methods for monitoring drivers' attention. 20000 images of 100 different drivers were collected in real driving environments. The ratio of dynamic to static driving environment images is around 3:7. Data were captured under different weather conditions (sunny, cloudy, rainy, snowy), and during both day and nighttime from different kinds of vehicles (sedans, SUVs and trucks). The drivers' ages ranged between 20 and 70. The capture locations included city, highway and suburban. Yang *et al.* [34] have proposed a multi-task learning CNN framework with joint parameter sharing named DANet. They have proposed a Dual-Loss Block to decompose the pose estimation task into pose classification and coarse-to-fine regression. They have also introduced an Object-centric Aware Block that enables the prediction of gaze angle from predicted head pose and the relative position of the face to reduce orientation estimation errors. The mean absolute error (MAE) of their proposed method on the FDUD dataset is 3.72.

2) Studies Using Private Datasets: Murphy *et al.* [20] have proposed a new procedure for static head pose estimation and

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visual 3D tracking. In real-time, their system continuously detects and tracks drivers' head position and orientation with six degrees of freedom. Their system uses a localized gradient orientation histogram as input to support vector regressors. Their tracking module provides a fine estimate of the 3D motion of the head using a new appearance-based particle filter for 3D model tracking in augmented reality. Their reported MAE for head pose estimation is as follows: Pitch - 8.57, Yaw - 11.24, Roll - 8.29.

Tawari *et al.* [25] used three cameras (mounted facing the driver on the left, front and right side of the wheel) to track facial features, analyze their geometric configuration, and estimate head pose using a 3D model. Inertial measurement unit (IMU) sensors placed on drivers' heads and fixed at the back of the car were used to track their respective motions for evaluation. Either a Constrained Local Model or Mixture of Pictorial Structures (MPS) was used for feature extraction. Using four features, extracted points are reconstructed on a 3D model, with the leftmost camera initialized for tracking and transitions to others to gain better perspective. With this three-camera framework, the researchers' MPS method resulted in a lower error rate when detecting the yaw angle (within 0-20 degrees) of drivers' head pose.

Yu *et al.* [72] focused on face tracking. They collected their own data at a resolution of 800×600 pixels using a digital camera. They designed a real-time embedded face tracking system on a single FPGA chip. They have proposed a method based on semantics-based vague image representations (SVIR) [73]. They track the face by the eyes, and each eye's SVIR features are kept in the FPGA's register to save on the computing load.

Jing *et al.* [74] also focused on the face detection task. They built their own dataset, which contains 1680 driver face images collected from the internet. They have proposed a method using a cascaded CNN, which contains multitask branches, including face classification, bounding box regression and the detection of ten key facial points. The performance of their model was evaluated by the precision, recall and F-measure metrics, which were reported to be 94.382%, 86.921% and 90.517%, respectively.

Liu *et al.* [75] synthesized virtual driver data by using a computer animation engine. This virtual dataset was combined with a small real-driver dataset to train their model in a transfer learning manner. Their real data (target domain) contain 8 actors and 4,152 frames captured by a panoramic front-view IR camera with a fisheye lens. The researchers labeled each frame for both the driver and the passenger sitting in front. Next, 323,740 virtual data (source domain) frames with 280 actor models were synthesized and then labels were generated automatically by the animation engine. Their proposed RM-ThinNet was reported to achieve superior detection accuracy in real-time (81 fps) for multiple tasks. More specifically, 91.2% (<5 pixels) for 9-joint body detection, 92.1% (<4 pixels) for 5-point face detection, and 83.3% (<5 degrees) for head angle estimation. Compared to ResNet [58]- and DenseNet [59]-based methods, RM-ThinNet is 5-12 times smaller in size and 8-25 times faster in speed.

This survey's review of the literature suggests that face detection has become a well-studied concept over time. Using datasets such as SHRP2, VIVA and FDUD, a range of approaches from traditional methods to deep models have been employed to detect drivers' faces with high accuracy. The face detection task serves as a stepping stone for following procedures to extract features or high-level information about driving tasks. Various methods have been developed to address specific challenges in detecting a driver's face, such as occlusion due to weather or insufficient illumination. Also, moving past detecting from stationary images, efforts are underway to develop deep model-based systems to detect drivers' faces in real-time, which has potential to be highly useful in autonomous vehicles and the development of driver assist safety systems.

III. High-Level Tasks

The classes in the high-level category (namely detecting drowsiness, distracted behavior and lane changes) cover important scenarios for driver analysis. Distraction and drowsiness have been highlighted as contributing factors to unsafe driving behavior responsible for traffic crashes. Drowsiness detection is a binary classification problem. Drowsiness can be detected by analyzing related physical and physiological changes through visual and nonvisual features. Visual features are captured by cameras, and nonvisual features are extracted from sensors that record physiological data such as EEG output. Distracted behavior is defined as a driver interacting with objects other than the steering wheel, such as using a cellphone, eating, putting on makeup, and talking to passengers, when this leads to taking one's eyes off of the road.

A. Drowsiness

Driver drowsiness is a major reason for road accidents. According to the National Highway Traffic Safety Administration (NHTSA) [76], there are a total of approximately 8,000 deaths attributed to drowsy driving each year. Hence, it is imperative that the research community understand and pursue efforts to minimize accidents due to drowsiness. This section summarizes studies focusing on detecting driver drowsiness, together with the dataset(s) they have used and their performance. This analysis will focus on three public and nine private datasets collected by various researchers. A drowsiness detection task is defined in this review as detecting a drowsiness feature such as yawning or eye closure. This task is the same across public and private datasets, with the only difference being that private datasets are more feature-rich than public datasets. For image-based datasets, both public and private datasets have only used a single-camera setup positioned around the rearview mirror. This placement helps to fully capture a driver's changes in facial expression but not block the driver's field of view.

1) Studies Using Public Datasets: There are several public datasets for drowsiness detection that have only recently been released. Among these, we have chosen to focus on three public datasets, namely YawDD [38], DROZY [39], and NTHU [40], which are commonly used in the research

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TABLE VII
SUMMARY OF REVIEWED PUBLIC DATASETS USED FOR DROWSINESS DETECTION

	YawDD [38]	DROZY [39]	NTHU [40]
Year	2014	2016	2016
FPS	30	30	15/30
Resolution	640x480	512x424	640x480
No. of videos	342	500K frames	360
No. of participants	107	14	36
Sensor Type	RGB	ECG/EEG/ EOG/EMG/IR	IR/RGB

community to benchmark the drowsiness detection task. These datasets are summarized in Table VII, and studies analyzing them are discussed below, and summarized in Table VIII.

- **YawDD** was presented by Abtahi *et al.* [38]. This dataset contains 342 videos collected with Canon A720Is from 107 individuals. Videos were collected with a resolution of 640 × 480 and 24-bit true color RGB) at 30 fps. There are 57 male and 50 female participants with varying ages, ethnicity, and facial characteristics. There are also varying scenarios to diversify the dataset, such as participants with/without glasses, with/without facial hair, and different hairstyles and clothing. The camera is roughly located at the rear-view mirror, as seen in Fig. 2. After face detection, it is important to extract the features of mouth or eyes, which are commonly used as parameters for driver drowsiness. Zhang *et al.* [77] have proposed a yawn detector by using a nose tracking and face motion orientation module as inputs along with cropped frames of the driver's mouth. The nose tracking module uses Kalman Filter to estimate the search area based on the previously detected nose location. Their model obtained 92% accuracy with 13% false positive rate. Akrout and Mahdi [78] have proposed a yawn detector model using lips surface evolution and the internal zone of the mouth as a spatio-temporal descriptor. They first perform face and mouth detection using the Viola-Jones algorithm. Then, the lips extraction step is performed to extract the external contour of the lips. Finally, the spatio-temporal descriptor was obtained using the detected lip region from the previous step, and the internal zone of the mouth. This model obtained a yawning detection accuracy of 83% and recall of 82%.
  Besides traditional methods, studies deploying deep learning models have become popular. Zhang and Su [79] proposed a yawn detector based on a spatio-temporal approach using CNN and Long Short Term Memory (LSTM) modeling. They used GoogLeNet, pre-trained on ImageNet dataset and State Farm DDD dataset, before fine-tuning on YawDD. On top of GoogLeNet, this model uses three layers of stacked LSTM, and a softmax classifier as the final layer. This method obtained 87.1% and 88.6% for recall and accuracy rate of yawn detection, respectively. Saurav *et al.* [80] proposed a Bi-Directional LSTM with CNN features as the drowsiness detection model. The pipeline first detects the face and mouth, followed by cropping the mouth region to a 48×48 size, which is then fed into a CNN feature extractor. Features of 32 frames are stacked and passed into a Bi-LSTM layer with a final sigmoid layer to classify whether they correspond to a yawn or not. The authors reported a validation accuracy of 96.48%.
  Eyes are an important indicator for drowsiness accounted for in several studies. Miah *et al.* [82] proposed a method to detect drowsiness by measuring mean eye landmark distance, along with closed eye duration. They first detected faces and eyes, followed by detecting 68 facial landmark points. Then, left and right eyes were cropped for mean eye landmark computation. They achieved 93% precision in drowsiness detection. While the discussed studies perform drowsiness detection on collected datasets, the future of driver assist systems requires a real-time model capable of alerting the driver. You *et al.* [81] proposed a real-time fatigue detection model based on YOLOv3-tiny [89]. They introduced a geometric area called the Face Feature Triangle to capture the driver's facial motion state to create a Face Feature Vector (FFV). A sliding window approach was used over the FFV to get facial information entropy. This approach achieved a driver fatigue prediction accuracy of 94.32%.
- **DROZY dataset** was presented by Massoz *et al.* [39]. This dataset contains infrared videos and polysomnography PSG) data of participants undergoing the psychomotor vigilance test (PVT). Infrared videos were collected with a Microsoft Kinect v2 sensor, while PSG signals such as electroencephalogram (EEG), electrooculogram (EOG), electrocardiogram (ECG) and electromyogram (EMG) were collected with an Embla Titanium system sampled at 512 Hz. Infrared videos have a resolution of 512×424 and are 16-bit. They were collected at 30 fps. The infrared camera was placed around 0.7m away from the participants, right below the direct line of sight as seen in the third row of Fig. 2. There is a two-step annotation process, manual and automatic annotation. They manually selected 720 frames from 14 participants to ensure a diverse appearance distribution. Maftukhaturrizqoh *et al.* [83] proposed to detect drowsiness using a radial basis function network as a classifier with electrocardiographic RR statistical feature. This work starts by performing feature extraction into 12 statistical RR interval features: mean, variance, standard deviation, geometric mean, harmonic mean, interquartile, kurtosis, skewness, median, mode, maximal, and minimal. These features were fed into a neural network classifier. The reported testing accuracy on the DROZY dataset is 81.96%. Krishnan *et al.* [84] proposed a drowsiness detection model using a Support Vector Machine (SVM) and K-Nearest Neighbour (KNN) as classifiers. They used the framing method to segment the signals into equal frames of 2 seconds using a rectangular window with 50% overlap. Five EEG channels from the raw multimodal signal were filtered through a band-pass filter at a cut-off frequency of 50 Hz. The log energy entropy and band power features were extracted from each frame, then rescaled using bipolar normalization

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TABLE VIII
SUMMARY OF METHODS USING PUBLIC DATASET FOR DROWSINESS DETECTION

Paper	Year	Dataset	Method	Focus	Performance
Zhang et al. [77]	2015	YawDD	MLP utilizing nose tracking & mouth region	Yawn	92% accuracy, 13% FPR
Akrout and Mahdi. [78]	2016	YawDD	Zero-crossing	Yawn	83.0% accuracy, 82% recall
Zhang and Su [79]	2017	YawDD Own Dataset	CNN-LSTM with canny edge augmentation	Yawn	87.1% recall, 88.6% accuracy 85.3% recall, 86.0% accuracy
Saurav et al. [80]	2019	YawDD	CNN with Bi-Directional LSTM	Yawn	96.48 val acc%
You et al. [81]	2020	YawDD	SVM with facial motion information entropy	Facial fatigue motion	94.32 acc%
Miah et al. [82]	2020	YawDD	Mean eye landmark distance	Blink	93.0 % precision
Mafukhaturriqoh et al. [83]	2019	DROZY	Radial Basis Function Network with Electrocardiographic RR Interval	ECG Signals	81.96% acc
Krishnan et al. [84]	2020	DROZY	Electroencephalogram Anomaly based on Spectral Entropy	ECG Signals	95.2% acc, 96.7% sensitivity, 94.2% specificity, 93.5% TPR, 40.2% TNR
Yu et al. [85]	2018	NTHU	3D CNN with condition-adaptive representation learning	Yawn	76.2% acc
Imranul et al. [86]	2019	NTHU	MLP and SVM using HoG & Gabor Features	Yawn	HoG — SVM: 88.04% acc, NN: 91.21% acc Gabor — SVM: 86.75% acc, NN: 89.63% acc
Vu et al. [87]	2019	NTHU	Gated-based RNN & Voting Layer	Yawn	84.81% acc
Ghourabi et al. [88]	2020	NTHU	MLP and K-NN utilizing EAR & MAR & Optical Flow	Yawn, Blink, Nod	74.91% average F-score

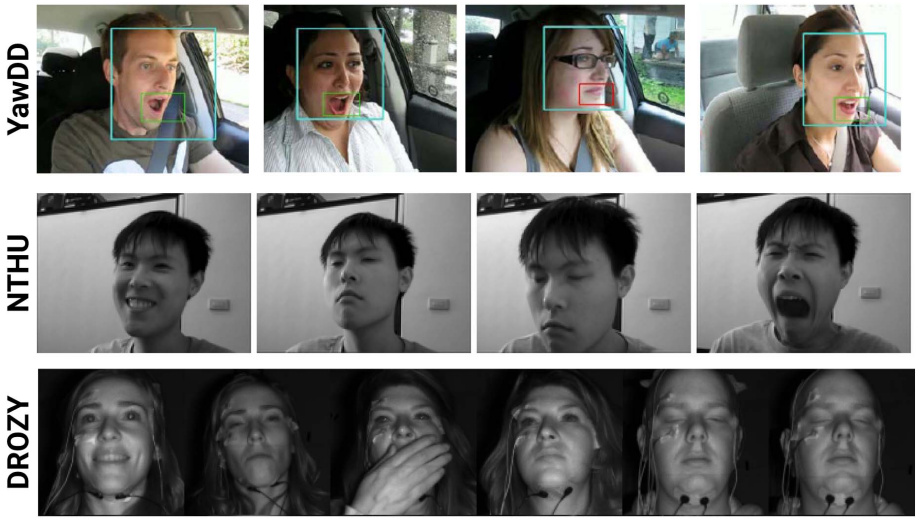


Fig. 2. Visualization of sample frames from each respective public dataset in this section. (Top) YawDD dataset [38], (Middle) NTHU dataset [40], (Bottom) DROZY dataset [39].

before being used as inputs to train the classifier. This paper achieved the best accuracy of 95.2% via KNN. Other performance metrics were reported as well, such as sensitivity (96.7%), specificity (94.2%), true positive rate (93.5%), and true negative rate (40.2%).

- **NTHU dataset**, which was presented by Weng *et al.* [40], contains 360 infrared and regular videos. The infrared videos were recorded with a D-Link DCS-932L at a resolution of 640 × 480. The regular videos were recorded with Logitech C310 HD with a resolution of 720p and 24-bit true color (RGB) at 30 fps. There are 36 participants with varying age, ethnicity, skin color, and gender. Each participant is instructed to perform actions under 5 scenarios: BareFace, Glasses, Sunglasses, Night-BareFace, and Night-Glasses. Yu *et al.* [85] have proposed a conditional-adaptive representation learning based on the 3D-deep convolutional network. Their model consists of four parts: spatio-temporal representation learning, scene understanding, feature fusion, and drowsiness detection. The spatio-temporal module extracts spatio-temporal features from multiple consecutive frames to discover rich and discriminative features. The scene understanding module interprets driver or environmental elements such as movement of facial features and time of day. The feature fusion model then learns a conditional-adaptive representation given extracted spatio-temporal and scene understanding features. Finally, the drowsiness module outputs a binary output classifying if drowsiness is detected or not. They reported F-scores of 76.5% and 76.0% for drowsiness and non-drowsiness, respectively, and obtained an accuracy of 76.2%.

Vu *et al.* [87] proposed a driver drowsiness detection approach using CNN, Gated-based Recurrent Neural Networks (ConvCGRNN), and a voting layer. They perform facial detection and tracking, followed by passing a cropped face window of 128 × 128 into a CNN to extract facial features. A total of 3 seconds of frames are extracted and passed into a ConvCGRNN layer, followed by a voting layer independently classifying each frame.

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Each voting layer acts as a classifier, and ultimately the results of the drowsiness detection were decided by the maximum value of the voting outputs. The proposed model obtained an accuracy of 84.81%, with a drowsiness and non-drowsiness F-score of 86.28% and 82.99%, respectively, across all driving scenarios.

Imranul *et al.* [86] proposed detecting drowsiness by combining eyes, face, and mouth features through HOG and Gabor filter. These features are then concatenated for two classification methods, SVM and a Neural Network (NN). For HOG features, the performance metrics for SVM is 88.6% (Drowsy), 87.48% (non-drowsy), and 88.04% (Accuracy). Likewise, the performance metrics for NN, using HoG features, are 91.98% accuracy on drowsiness detection, 90.45% accuracy on non-drowsiness detection, and overall accuracy of 91.21%. For Gabor features, SVM performed slightly worse, obtaining accuracies of 87.45%, 86.5%, and 86.75%, for drowsy, non-drowsy, and average overall detection, respectively. For NN using Gabor features, the performance also dropped compared to NN using HOG features. The NN model using Gabor features obtained drowsiness detection accuracy of 91.25%, non-drowsy detection accuracy of 89.12%, and overall accuracy of 89.63%.

Ghourabi *et al.* [88] used a joint monitoring approach to detect drowsiness, specifically monitoring eye blinks, yawning, and nodding. They used HOG to detect faces, since HOG can detect faces in real-time and has shown to be robust against different head poses and lighting conditions. A standard set of 68 facial landmarks was placed after the face detection. They used facial landmarks around the eyes and mouth to compute Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR). Nodding was monitored by computing the optical flow of the head movement, measured by first determining the central location of the face, followed by computing the optical flow using Lucas- Kanade's algorithm. The model obtained a drowsiness detection F-score of 79%, non-drowsiness F-score of 70.83%, with an average F-score of 74.91%.

Numerous studies have been conducted using both the YawDD and NTHU datasets as compared to the DROZY dataset. We therefore infer that the majority of work done using public datasets has been image-based detection as compared to sensor-based detection. For image-based detection, most researchers have focused on detecting yawns [77]–[80], [85]–[87] as compared to eye blinks [82], facial fatigue motion [81], or combination of these [88]. The best reported performance using the YawDD dataset utilized Bi-Directional LSTM [80]. The second best reported performance did not use LSTM, but instead utilized temporal information as a sliding window to capture entropy information across time [81]. For the NTHU dataset, the best reported performance utilized the HOG features of input images to pass them into a Multi-layered Perceptron (MLP) [86]. The second-best reported performance used a gated-based RNN with a cumulative vote approach to classify drowsiness frames [87]. From these papers, it appears that utilizing temporal layers improves the performance of drowsiness detection.

2) Studies Using Private Datasets: The papers in this section employed private datasets to detect drowsiness. All the papers covered in this section employed one camera facing the driver except the work in [95], wherein, in addition to the data from a face-view camera, the movements of the pedals and steering wheel, driver's electrocardiogram and respiration, and the seat pressure distribution was measured. In general, drowsiness detection has been performed by using eye blink rate and/or head position and by employing different types of classifiers, such as neural networks, SVM, thresholds, random forests etc. Different methods that have used private datasets for drowsiness detection are listed and summarized in Table IX. Models in [91], [93]–[95], [98] defined a ground truth level of drowsiness, while the models in [90], [92], [96] and [97] did not. Instead, these latter models used low-level metrics to numerically evaluate the performance.

Deng *et al.* [96] have presented a system to identify drivers' fatigue by detecting yawning, blinking and eye closure duration. They introduced an improved KCF face-tracking algorithm named MC-KCF. They fused FHOG features with CNN extracted features to provide more reliable and robust face tracking results. They used Dlib [69] to locate 68 facial key points and identify regions of eyes and mouth. A simple CNN with fully connected (FC) layer is used to classify the fatigue state. They reported 95% face tracking accuracy and 93.6% eyes closure and yawning detection accuracy. Arslan *et al.* [97] and Deng *et al.* [96] employed CNN-based classifiers to detect eye states. Generally speaking, they first used a face detector to extract the facial features of eyes and/or mouth, and then used a CNN to detect eye closure and/or yawning. More specifically, Arslan *et al.* [97] proposed a real-time drowsiness detection system, using a camera facing the driver's face. Their proposed method used face and eye features with eye blinks to detect drowsiness. Viola-Jones algorithm [68] was used to detect eye regions, which are then cropped and sent to a CNN-based network for eye state detection. They reported 97.8% accuracy in detecting eye state being open or closed on their private dataset, which includes a total of 4,000 frames.

There have also been other methods, which are not CNN-based, but instead use classifiers such as SVM, HMM, etc. together with handcrafted features. Mbouna *et al.* [91] used visual features, such as eye index, pupil activity and head pose, together with an SVM to classify the level of drowsiness. The eye index measures eye closures; pupil activity finds dynamic motion of the eye; and head pose calculates all directional head movements. The three types of visual features were extracted from each video frame, and then an SVM classifier was used to identify the drowsiness level. They collected 1,521 frames, and obtained the drowsiness ground truth by experts voting. They reported 94.84% accuracy for drowsiness level detection. Choi *et al.* [93] proposed a PE-ASM, which is an extension of Discriminative Bayesian Active Shape Model (DB-ASM) [99], to extract two facial features (head pose and eye-blinking) by incorporating six extreme head pose cases. For drowsiness

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TABLE IX
SUMMARY OF METHODS USING PRIVATE DATASETS FOR DROWSINESS DETECTION. THE NUMBER OF FRAMES COLLECTED FOR EACH DATASET IS LISTED IN THE THIRD COLUMN

Paper	Year	# of Frames	Classifier	Focus	Performance Metrics
Sigari et al. [90]	2013	136K	Adaptive Fuzzy	Head pose & Eye closure	90.8% precision 7.7% FPR,15.4% FNR
Mbouna et al. [91]	2013	1.5K	SVM	Drowsiness	94.84% accuracy
Wongphanngam et al. [92]	2016	160	Random Regression Forests	Head pose	93.75% sensitivity
Choi et al. [93]	2016	23K	Hidden Markov Model	Drowsiness	92% accuracy
Baek et al. [94]	2018	5K	AdaBoost	Drowsiness	87% accuracy
Sunagawa et al. [95]	2019	N/A	RBF-kernel based support vector regression	Drowsiness level	0.599 RMSE
DriCare [96]	2019	1M	MC-KCF CNN	Face tracking & Eye closure	95% accuracy/93.6% accuracy
Arslan et al. [97]	2020	4K	CNN	Eye closure	97.8% accuracy
Sankaran et al. [98]	2020	N/A	PERCLOS	Drowsiness	80% precision

detection, they used two Hidden Markov Models to identify whether a driver is drowsy or not with 92% accuracy.

Wongphanngam *et al.* [92] proposed a driver fatigue warning system using Kinect depth images. They first converted a depth image into gradient image to detect the driver's face, and then used Discriminative Random Regression Forests to estimate the angles of head rotation. The system would warn the driver by monitoring the pitch angle of head orientation to detect a nodding off event. They reported 93.75 sensitivity on their private dataset, which includes 160 images from 40 subjects. Baek *et al.* [94] used an AdaBoost classifier based on Census Transform features to detect the driver's face, and then found the face landmarks, and calculated the percentage of eye closure (PERCLOS) for drowsiness detection. They reached 87% average accuracy over five subjects by using PERCLOS-10 model on their private dataset, which contains 5,000 samples. Sankaran *et al.* [98] used Viola-Jones algorithm to detect faces and eyes, and calculated PERCLOS. They used a simple threshold on PERCLOS to classify drowsiness level, and reached a precision of 0.8. Sunagawa *et al.* [95] presented a drowsiness level detection method. They collected their own dataset in a virtual driving environment from 49 participants (average age: 38.8 ± 5.7; 26 males and 23 females). A face-view camera was used to record participants' faces at 60 fps. In addition, the driving simulator recorded the movements of the pedals and steering wheel at 10 Hz. The subject's electrocardiogram and respiration were measured at 200 Hz, and the seat pressure distribution was measured at 5 Hz. They introduced posture index, and the overall drowsiness index includes {vehicular index, blink index, posture index, physiological index}. They employed Radial Basis Function (RBF)-kernel based-support vector regression [100] to calculate the drowsiness level. Root mean square error (RMSE) was used to evaluate the performance, and the lowest RMSE was reported to be 0.599, compared to the RMSE of 0.91 of the baseline method [101].

Among these private dataset studies, researchers have utilized more traditional machine learning approaches than deep learning approaches to perform drowsiness detection. However, both top-reported performances have come from studies that utilized deep learning approaches [96], [97]. The next two closest performing methods, which are not based on deep learning, used SVM [91] and Random Regression Forests [92].

TABLE X
SUMMARY OF THE PUBLICLY AVAILABLE DATASETS FOR DISTRACTED BEHAVIOR DETECTION AND CLASSIFICATION

	SHRP2 [33]	StateFarm [30]	AUC [31]
Year	2015	2016	2017
No. of frames	91K	22K	17K
No. of views	5	1	1
No. of participants	41	26	31
Male/Female	N/A	N/A	22/9
Num. of classes	6	10	10
Resolution	356x240	640x480	1920x1080
Camera type	GRAY/IR	RGB	RGB

B. Distracted Behavior

Distracted driving is a major source of injury and death for motorists in the United States [102]. In this section, we introduce papers that focused on detecting and classifying distracted behaviors by using sensors such as camera, accelerator, wheel info, and ECG. We first introduce papers that used public datasets followed by papers that used self-collected private datasets.

1) Studies Using Public Datasets: Four publicly available datasets for distracted behavior detection and classification are listed and compared in Table X, and described in more detail below. The SHRP2 [33] dataset contains the largest number of participants, which is 41; however its image resolution is the lowest. The StateFarm [30] dataset's image resolution is also relatively low and this dataset does not provide sequential information. The AUC [31] dataset has the highest resolution.

- **StateFarm Distracted Driver Dataset (DDD) [30]** is a publicly available dataset, which is commonly used as a benchmark for the recognition of distracted driver behavior. It contains 22,424 image frames, with a 640×480 resolution, from 26 participants. Images were captured by a visible range (RGB) camera providing a side-view of the driver. Data were captured from four cars and cover the following 10 classes: c0: normal driving, c1: texting-right, c2: talking on the phone-right, c3: texting-left, c4: talking on the phone-left, c5: operating the radio, c6: drinking, c7: reaching behind, c8: hair and makeup, c9: talking to passenger. Since this dataset does not provide sequential information, most papers have

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TABLE XI
COMPARISON OF SINGLE-FRAME-BASED MODELS ON THE STATEFARM DISTRACTED DRIVER DATASET (DDD)

Paper	**Year**	**Method**	number of classes	**Accuracy**
Torres et al. [114]	2017	PCN and YOLOV3	2 (using phone or not)	99.68%
Jegham et al. [18]	2018	KNN	10	70%
Valeriano et al. [103]	2018	ResNet101	10	82.38%
Unadkat et al. [105]	2018	AlexNet	10	99.49%
Masood et al. [106]	2018	VGG16	10	99.57%
Baheti et al. [104]	2018	Modified VGG16	10	95.54%
Varaich et al. [108]	2019	Inception v3	10	99.06%
Rao et al. [109]	2019	PCA + CNN	10	97.31%
Wang et al. [37]	2019	Attention-ResNet50	10	92.4%
Mofid et al. [110]	2020	Preprocessing + ResNet50	10	66.2% (F1 score)
MSA-CNN [112]	2020	Attention+CNN	10	91.8%
Dhakate et al. [111]	2020	Ensemble	10	97.00%
Huang et al. [113]	2020	Ensemble	10	96.74%

proposed single-frame-based classification methods [18], [37], [103]–[113]. Single-frame-based models classified the driver's behavior for each frame without incorporating temporal information. Although the neural network structures vary, the vast majority of the proposed methods used an end-to-end CNN-based classifier [105], [106], [108]. Different papers employed different base models to extract features, and some of them introduced additional attention mechanisms [37], [112] or pre-processing steps [109], [110] to improve the quality of features. Another popular approach is ensemble training [111], [113], which uses multiple CNN networks to extract features from the same input frame and an ensemble of those features together for the final classification. A comparison of different single frame-based models is provided in Table XI, and these models are described in more detail below.

Torres *et al.* [114] focused on detecting mobile phone use by employing two steps. They performed face tracking by using PCN [115], to provide the detection area, and then used YOLOV3 [116] to detect a cellphone in that area. The reported accuracy was 99.68%. Unadkat *et al.* [105] trained two models based on AlexNet [117] and Inception-V3 [118]. Masood *et al.* [106] used VGG16 and VGG19 pretrained on ImageNet. Baheti *et al.* [104] proposed a modified VGG16 using LeakyReLU [119] instead of ReLU. They also added a batch normalization layer to the original VGG16 model [120], and replaced the dense layer of VGG16 with 1×1 convolutions, which greatly reduced the number of parameters. Varaich *et al.* [108] used ImageNet-pretrained Inception-V3 and Xception [121]. Rao *et al.* [109] used PCA to preprocess the images, and then used a CNN network with a dense layer as the classifier. Similar to Rao [109], Mofid *et al.* [110] also proposed a preprocessing method. They used several different preprocessing and augmentation techniques, including random brightness adjustment, random rotation adjustment, facial blurring and skin segmentation. They concatenated the augmented data as the input and used Resnet-50 [61] as the classification model. Jegham *et al.* [18] first segmented body parts using SURF key points. Then, they extracted HOG features on candidate regions to do classification by using KNN. Hu *et al.* [112] proposed a multi-scale CNN model with attention mechanism to learn pixel saliency and channel saliency. Wang *et al.* [37] proposed an attention module, inserted it into VGG16 and ResNet-50, and improved both of their performances. Dhakate and Dash [111] and Huang *et al.* [113] created an ensemble of features from different CNN models, including ResNet-50, Xception, Inception-v3 and VGG19, by concatenation.

By manually adding the sequential information among frames, the StateFarm DDD dataset was also used in models incorporating temporal information [103], [122]–[125]. These models share a similar structure of extracting features from each frame by a CNN and learning from the sequential features for the final prediction. The comparison of these models is provided in Table XII, and they are described in more detail below.

Valeriano *et al.* [103] investigated both single-frame- and sequential-frame-based models. For the single-frame-based model, they used ResNet-101 pretrained on ImageNet, and fine tuned it on this dataset. For the sequential-frame-based model, they used the optical flow branch of I3D [126] pretrained on the Kinetics dataset [127]. Moslemi *et al.* [122] also used two-stream I3D model, which was pretrained on both the ImageNet and Kinetics datasets, and then fine-tuned on this dataset. Kose *et al.* [123] calculated the optical flow of the input data and concatenated vertical flow and horizontal flow to the RGB input, which they referred to as data level fusion. They sparsely selected N frames from action video, used a CNN to extract the corresponding features and fused them together for final classification. Li *et al.* [124] proposed a spatio-temporal driver action recognition model using skeleton-based pose estimation approach. Given a side view of the driver, the driver's pose was estimated to obtain joint positions as the input for a Graph Convolutional Network. Alotaibi *et al.* [125] proposed a combination model using Inception [118], Hierarchical

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TABLE XII

COMPARISON OF THE MODELS, WHICH INCORPORATE TEMPORAL INFORMATION, ON THE STATEFARM DISTRACTED DRIVER DATASET (DDD)

Paper	Year	Method	Accuracy
Valeriano et al. [103]	2018	OpticalFlow + 3D CNN	92.74%
Moslemi et al. [122]	2019	two stream + 3D CNN	94.4%
Kose et al. [123]	2019	BN-Inception	99.10%
Li et al. [124]	2019	keypoints+GCN	88.2%
Alotaibi et al. [125]	2019	Modified inception	92.23%

Recurrent Neural Network [128], and ResNet [61] to build a distracted driver classification model. On this dataset, most of the papers have used a CNN model. Torres *et al.* [114] reported the highest accuracy which was 99.68%, however, they only focused on mobile phone detection. Among the papers, which focused on all the classes in the dataset, Masood *et al.* [106] reported the highest accuracy of 99.57%, and the model they used was VGG16.

- **American University in Cairo (AUC) [31]** dataset contains 17,307 video frames with 1080×1920 resolution. Images were captured by one visible range (RGB) camera providing a side view of the driver. Data were captured from four cars with 31 participants, and covers the following 10 classes: drinking, drive safe, reach behind, talk left, text left, adjust radio, hair and makeup, talk to passenger, talk right, text right. This dataset was published by Abouelnaga *et al.* [31] and viewed inputs as single frames. They proposed a genetically-weighted ensemble CNN structure, and found that the ensemble of two AlexNet models resulted in a satisfactory real-time performance.

  Behera *et al.* [129] proposed a context-driven Multi-stream LSTM, where contextual information, such as body pose and body-object descriptors, were utilized as high-level representations to detect fine-grained driver activity. Arefin *et al.* [130] proposed a robust method, which consisted of a modification of AlexNet architecture and the aggregation of HOG features [131]. The number of parameters in the proposed model reduced from 62.3M to 9.7M compared to AlexNet. Eraqi *et al.* [132] further refined their approach by including skin segmentation into the ensemble of distracted driver classifier. They showed that they could train a smaller model, utilizing this ensemble method, which runs in real-time. Behera *et al.* [133], inspired by human perception, considered human activity as a combination of pose and semantic contextual cues. They modelled a pairwise relation to capture the structure information of body joints. They proposed a multi-stream network combining high-level semantics from different parts by classifier-level feature fusion. Mase *et al.* [134] utilized CNNs and stacked bidirectional LSTM to correlate spectral-spatial features to the correct body posture. Behera *et al.* [129] and Mase *et al.* [134] used video-based methods, while the other papers are frame-based. Abouelnaga *et al.* [31] reported the highest accuracy, which was 95.98%. Table XIII summarizes these studies using the AUC dataset.

  Roitberg *et al.* [135] introduced the concept of open sets to the area of driver observation, where methods have been evaluated only on a static set of classes in the past. To formulate the problem of open set driver activity recognition, a model was designed to identify behaviors previously unseen by the classifier. A combination of current closed set models and multiple strategies were adopted for novelty detection. The proposed method employed the prominent I3D architecture extended with modules for assessing its uncertainty via Monte-Carlo dropout. The problem was defined in two parts as novelty detection and multi-class open set recognition. Compared to baseline models, such as Random Chance, One-class SVM, Gaussian Mixture model and conventional neural networks, their proposed Bayesian I3D model achieved an ROC of 85.30% for the binary task of identifying classes unseen. In the multi-class open set recognition part, the authors' model classified with a balanced accuracy of 57.55%, which was better than the accuracy of conventional methods. The results represent clear benefits of uncertainty aware models for monitoring real-life driving activity.

TABLE XIII

COMPARISON OF DIFFERENT STUDIES USING THE AMERICAN UNIVERSITY IN CAIRO (AUC) DATASET

Paper	Year	Method	Accuracy
Abouelnaga et al. [31]	2017	genetically-weighted ensemble CNN	95.98%
M-LSTM [129]	2018	Multi-stream LSTM	86.25%
Eraqi et al. [132]	2019	genetically-weighted ensemble CNN	90%
Arefin et al. [130]	2019	AlexNet+HOG	93.65%
Mase et al. [134]	2020	CNN-BiLSTM	92.70%
MDFN [133]	2020	Modified VGG	95.54%

TABLE XIV

COMPARISON OF DIFFERENT STUDIES USING THE SHRP2 DATASET

Paper	Year	Method	Accuracy (cell phone usage	Accuracy (hand on wheel)
Seshadri et al. [136]	2015	Real AdaBoost	93.9%	NA
Hoang et al. [137]	2016	MS-FRCNN	94.2%	93%
Le et al. [138]	2017	CNN&RCNN	90%	82.1%
Ulhaq et al. [139]	2017	SVM&RCNN	93.2%	NA

- **SHRP2 Dataset** was described in Section II-B. Studies using the SHRP2 dataset for distracted behavior detection are mainly based on object detection models, and they all focus on detecting cell phone use and/or detecting hands on the wheel. Table XIV shows a comparison of these works, and they are described in more detail below.

  Seshadri *et al.* [136] focused on detecting cell phone use by drivers. They used Supervised Descent Method to crop a region of interest. Classification was performed using Real AdaBoost, Random forest and SVM. The highest accuracy reported was from Real AdaBoost, which was 93.9%. Hoang *et al.* [137] proposed a Multiple Scale Faster-RCNN (MS-FRCNN) model to detect hands, face, steering wheel and the cell phone. Then, based on the

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location of these detected parts, they predicted if the driver was using phone, or if both hands were on the steering wheel using the last layer. The reported accuracy for cell-phone usage was 94.2%. The detection of hands-on wheel was separated into two tasks: task 1 was detecting if there were any hands on the steering wheel, and task 2 was detecting the number of hands on the steering wheel. The reported accuracy values for task 1 and task 2 were 93% and 65%, respectively. Le *et al.* [138] used a deep CNN to extract features of a driver, such as torso, seat belt etc. A probability map was drawn providing the confidence score for pixels to attribute to a particular object. RCNN was employed to extract features from bounding box regions, and an SVM classifier provided a score based on the extracted features. The reported accuracy values for hand on wheel detection and hand on phone detection were 82.1% and 90%, respectively. Ulhaq *et al.* [139] proposed a model that can detect cell phone usage from traffic control cameras using SVM and Spectral RCNN. They learned action specific deep action-lets, and used these activations to find regions of interest for faster RCNN, which performs the classification. The reported accuracy was 93.2%.

As seen in Table XIV, for cell phone use, the highest accuracy of 94.2% was reported by Hoang *et al.* [137]. For hand on steering wheel detection, the highest accuracy of 93% was reported by Hoang *et al.* [137] as well.

2) Studies Using Private Datasets: We have classified papers on distracted behavior detection that used private datasets into two categories: (A) studies using only camera(s); and (B) studies using a combination of cameras and other sensors.

a) Studies using only camera data: Studies using only the camera sensor modality have been further classified into studies using one camera, and studies using multiple cameras.

A.1 Studies Using One Camera:

- **A.1.i Face View:** The approaches in this category can be broadly classified into two classes: those focusing on the subject [140]–[142] versus focusing on the distracting object [107], [143]. The methods focusing on the subject analyze the facial features and distracted behavior of the driver directly. The methods focusing on the objects try to detect distracted behavior by detecting distracting objects. Studies using a single face-view camera are summarized in Table XV, and described in more detail below.

  Gao *et al.* [140] focused on drivers' facial features to detect driver distraction. They collected their own dataset with image resolution of 1280×1024 pixels. They used supervised descent method to extract facial features, and employed an SVM to classify the objectives such as emotional stress or head pose on two sets of data. The accuracy values reported on set1 and set2 were 87.1% and 81.5%, respectively. Xing *et al.* [141], [144] captured their own dataset, which covers the following seven classes: normal driving, right mirror checking, rear mirror checking, left mirror checking, using video device, texting, answering mobile phone. They used a GMM to extract a driver region, and then tried different CNN architectures, such as Alexnet, Googlenet and Resnet-50, to classify driver activities. The highest reported accuracy of 78.6% was obtained when AlexNet was used.

  Xiong *et al.* [107] focused on distraction due to cell-phones. Their dataset includes 47,680 images with the resolution of 1280×720 pixels. They trained a progressive calibration network (PCN) to detect the face and propose possible locations for the cellphone. Next, they used YOLO-V3 [116] to detect the cellphone location. The proposed and detected locations of the cellphone were compared to decide if the driver was using a cellphone. The reported accuracy was 96.56%. Mao *et al.* [143] focused on detecting smoking and phone usage. They used PCN (Progressive Calibration Network) and DSST (Discriminative Scale Space Tracking) algorithms for face detection and dynamic tracking of faces. They employed a pre-trained YOLOV3 model to detect a phone or cigarette in the face area. The precision of cigarette detection and phone detection were reported to be 95.7% and 97.3%, respectively.

  A.1.ii Side View: Studies in this category employ a single camera providing a side view of the driver [145]–[149], and use similar labels and camera setup as the StateFarm DDD [30] dataset. Earlier approaches [145], [150] used Random Forest for behavior detection by using handcrafted features. More recent methods [149], [151]–[153] have used CNN-based networks in an end-to-end fashion. The dataset properties and performance of these studies are summarized in Table XVI, and described in more detail below.

  Southeast University Driving posture Dataset (SEU) [145] contains 92,473 images collected by one RGB side-view camera from four cars and 20 participants. The image resolution is 640×480 pixels. The dataset covers the following six classes: call, eat, operate shift, hands on wheel, play cellphone, smoke. This dataset was used by [145]–[148]. Zhao *et al.* [145] used Random Forests and reported an accuracy of 90.63%. C. Yan *et al.* [146] first pre-trained a CNN model by an unsupervised feature learning method called sparse filtering [155]. They subsequently fine-tuned it for classification, and reported an accuracy of 99.78%. In another work of theirs [148], they proposed a spatio-temporal driving behavior model with a hierarchical classification strategy to train a shallow machine learning model. The reported accuracy was 90.62%. S. Yan *et al.* [147] used a Gaussian Mixture Model (GMM) [157] to extract skin like regions, and passed it to an R*CNN model [158]. The reported mAP was 97.76.

  Chen *et al.* [149] built their own dataset covering the following ten classes: safe driving, text right, phone right, text left, phone left, adjusting radio, drinking, reaching behind, hair or makeup, talking to passenger. They proposed a model including a spatial stream and a temporal stream. The input of the spatial stream is an RGB image obtained by averaging a sequence of frames on the temporal axis. The temporal stream input

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TABLE XV
COMPARISON OF STUDIES USING A SINGLE FACE-VIEW CAMERA FOR THEIR PRIVATE DATASETS

Paper	Year	Resolution	N classes	N frames	Method	Accuracy
Gao et al. [140]	2014	1280x1024	2	N/A	SVM	78.1%
Xing et al. [141], [144]	2018	N/A	7	N/A	GMM and AlexNet	78.6%
Xiong et al. [107]	2019	1280x720	2	47680	PCN and YOLOV3	96.56%
Mao et al. [143]	2020	N/A	2	N/A	PCN,DSST and YOLOV3	N/A

TABLE XVI
COMPARISON OF STUDIES USING A SINGLE SIDE-VIEW CAMERA FOR THEIR PRIVATE DATASETS

Paper	Year	Resolution	N Classes	N Frames	Method	Accuracy
Zhao et al. [145]	2012	640x480	6	92473	Random Forest	90.63%
Yan et al. [150]	2014	640x480	4	N/A	MHI, PHOG and random forest	96.56%
Gupta et al. [154]	2015	640x480	6	N/A	Newtons secant method	100
Yan et al. [146]	2016	640x480	6	92473	sparse filtering [155]	99.78%
Yan et al. [147]	2016	640x480	6	92473	R*CNN	97.76%(mAP)
Yan et al. [148]	2016	N/A	10	N/A	TV-L1 [156] and VGG16	68.25%
Tu et al. [151]	2017	N/A	5	36151	VGG19	75.3
Sun et al. [152]	2018	N/A	5	N/A	PMESH and proposed CNN	91.6%
Li et al. [153]	2020	N/A	5	106677	MLP and YOLO	92%
Chen et al. [149]	2020	N/A	10	N/A	TV-L1	68.25%

is the optical flow calculated by TV-L1 optical flow [156] method from input frames. These two streams were fused together by a fusion network for final prediction. The reported accuracy was 68.25%.

Li *et al.* [153] also captured their own dataset, which contains 10,6677 images and covers the following five classes: marker, touchscreen, phone, drink, text. They used YOLO to detect the driver's right ear and right hand, and then used MLP to take those regions of interest (ROIs) as input to predict corresponding distraction. The reported accuracy was 92%. Yan *et al.* [150] built a dataset similar to the SEU dataset. Their dataset covers four classes, namely operating the shift lever, talking on a cell phone, eating, smoking. Motion History Image (MHI) and Pyramid Histogram of Orientated Gradients (PHOG) have been used to train a driving Random Forest posture recognition model. The reported accuracy was 96.56%. Gupta *et al.* [154] captured their own dataset, which contains images with resolution of 640×480, and covers the following six classes: grasping the steering wheel, operating the shift lever, eating during driving, talking on a cellular phone, controlling the dashboard, sleeping while driving. They utilized skin segmentation to improve a CNN model for driver posture classification. The accuracy reported on their dataset was 100%.

Tu *et al.* [151] presented their own dataset, which contains 36,151 images, and covers the following five classes: call, drink, rest, talk, text. They proposed viewpoint warping to obtain a better action recognition model than non-viewpoint warping models. The dataset consists of slightly different viewpoints of a side-view image. They utilized a planar homography to warp an input image to a target image before training a CNN. The reported accuracy was 75.3%. Sun *et al.* [152] collected five driver activities, namely eating, calling, two hands driving, one hand driving and sleeping, in a simulated driving system. They proposed a driver posture recognition model using a prior model enhanced skin histogram method (PMESH). The reported accuracy was 91.6%.

A.2 Studies Using Multiple Cameras: The papers in this section used two or more cameras, which provided multiple views [159]–[161] and/or investigated different types of data (RGB and depth) [162]–[164]. How to fuse such multi-view data together is key. Using CNN-based networks is the main stream method to fuse features. In recent works [161], [164], researchers have also used LSTM to handle the temporal information between frames. Related papers in this category are summarized in Table XVII, and described in more detail below.

Two RGB cameras (one side view and one face view) were used in [159], [160]. Tran *et al.* [160] used individual frames as input and did not consider the temporal information. They used an MLP layer on top of a set of features from VGG16, Alexnet, Resnet and GoogleNet for behavior classification. Zhang *et al.* [159] collected their own dataset in a mock-up car environment. The resolution of the two cameras is 640×360 pixels, and the FPS of the video is between 17.14 and 27.47. The dataset includes nine different behaviors: normal driving, texting, eating, talking, searching, drinking, watching video, gaming and preparing. They proposed an Interwoven Deep Convolutional Neural Network (InterCNN) architecture. Their proposed InterCNN with MobileNet [165] convolutional blocks was reported in the paper as able to classify nine different behaviors with 73.97% accuracy.

Depth cameras are used in [162]–[165] to get distance information. Martin *et al.* [162] used two Kinect sensors, one providing a face view and another one providing view of hands. They extracted edge, color and texture information from the hand view, and head pose and landmarks from the face view. They then investigated semantic understanding of the driver's state based on SVM. Craye *et al.* [163] also used a Kinect sensor providing a face view. They extracted eye behavior (gaze and blinking), arm position, head orientation and facial expression, and merged them by the proposed

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TABLE XVII
COMPARISON OF STUDIES USING MULTIPLE CAMERAS AND THEIR OWN DATASETS

Paper	Year	Camera Facing	N Frames	Method	Accuracy
InterCNN [159]	2018	side view and face view	1.3M	CNN	73.97%
Tran et al. [160]	2018	side view and face view	200K	GoogleNet + MLP classifier	90%
Martin et al. [162]	2014	2 Kinect 1 hand 1 face	14K	SVM	91%
Deo et al. [161]	2019	face, hands, body pose, feet	234K	LSTM	MAE:0.449
Craye et al. [163]	2015	1 Kinect facing face	720K	AdaBoost	90%
Chuang et al. [164]	2019	2 Kinect facing driver	NA	LSTM	88%

TABLE XVIII
COMPARISON OF STUDIES USING CAMERAS TOGETHER WITH OTHER SENSOR MODALITIES AND THEIR OWN DATASETS

Paper	Year	Camera Facing	Sensor Data	Method	Performance
Darnet [166]	2017	side view	IMU	InceptionV3 and LSTM	Acc - 87.02%
Ou et al. [168]	2018	face and side view	Speed	CNN	Acc - 98.3%
Omerustaoglu et al. [167]	2020	side view	OBD of vehicle	CNN and LSTM-RNN	Acc - 85%

AdaBoost and Hidden Markov Model. Chuang *et al.* [164] used two Kinect sensors, which were mounted at both sides of the wheel to create a 3D point cloud of the driver. They extracted a skeleton and used an LSTM model to classify the driver's behavior. Deo *et al.* [161] collected data from four RGB cameras observing driver's face, hands, body pose and feet. In addition, a depth camera was used to provide a hand view to analyze the distances of the driver's hands from the vehicle control panel. They ran different CNN models to extract frame-wise features, and then used LSTM to predict driver's take-over readiness. They define a metric for the take-over readiness termed the 'Observable Readiness Index (ORI)'. The ground truth ORI values range from 1 to 5. Their model estimates the ORI (between 1 to 5) with an MAE of 0.449.

b) Studies using camera and other sensors: In addition to cameras, the studies in [166], [167], and [168] have used additional types of sensor data, such as gyroscope, accelerometer and GPS data, to analyze drivers' behavior. These studies are summarized in Table XVIII, and described in detail below.

Streiffer *et al.* [166] collected image data from an inside face-view camera, and Inertial Measurement Unit (IMU) data from a mobile device. They proposed a method using InceptionV3 and LSTM to provide the final results. Their model outperformed the baseline model [118] by achieving an accuracy of 87.02% on their own dataset. Ou *et al.* [168] proposed a danger level inference module combining the driver's head information, distraction activity information and speed information. They used the yaw and pitch to estimate the orientation of the driver's head and CNN to detect the distractions. The head information is extracted from the face-view camera by using the OpenFace [169] toolkit and distracted behavior was detected by a CNN network, which used ResNet-50 as the backbone. Speed information was directly read from a vehicle sensor. They used fuzzy logic rules as the danger level inference module based on those three types of information. For the distracted reorganization module, they reached 98.3% accuracy for normal driving, talking and texting. Omerustaoglu *et al.* [167] used additional sensor data collected from vehicle, including information such as acceleration, speed and throttle. Integrating the data from these sensors and video data, they created a two-stage system involving a vision-based CNN model and LSTM-RNN model to detect driver distraction. The inclusion of other sensor modalities improved the accuracy of the model by 9% to 85% while testing on their own dataset.

TABLE XIX
PROPERTIES OF BRAIN4CARS [32] DATASET

	Brain4Cars [32]
Year	2015
Num. of frames	2M
Num. of views	2
Num. of participants	10
Male/Female	N/A
Num. of classes	5
Resolution	1920x1088
Camera Type	RGB

C. Lane Change Prediction

Many accidents are inevitable since the reaction time for drivers is too short. Anticipating a driver's actions from recent or near recent driving videos has been an important topic for Advanced Driver Assistance Systems (ADAS). In this section, we cover recent studies focusing on lane change maneuvers. Research on lane change maneuvers involves predicting a lane change intention and detecting a lane change. Most of the existing studies (8 out of the 12 papers in this section) used the publicly available Brain4Cars dataset, and the rest of them used their own private datasets. Despite the various approaches presented in the literature, the vast majority share some or all of the following stages:

1) Feature extraction: features contain both in-vehicle features and environment features. In-vehicle features include the driver's head orientation and eye gaze, and are extracted from cameras watching the driver. Environment or outside features include lane state and vehicle dynamics extracted from outside cameras and vehicle sensors (GPS, accelerator, VBOX etc.)
2) Maneuver state: features from inside and outside have been used to predict the driver's lane change intention.

We will first present papers using a public dataset, and then describe papers using researchers' own private datasets.

1) Studies Using Public Datasets: In this section, we will summarize the papers using the publicly available Brain4Cars

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TABLE XX

MANEUVER PREDICTION PERFORMANCES ON THE BRAIN4CARS DATASET. OUTSIDE (NUMERICAL FEATURE) IS THE SENSOR DATA PROVIDED BY THE DATASET. FOR TIME-TO-MANEUVER (TTM) (SEC), LARGER NUMBER INDICATES BETTER PERFORMANCE

Paper	Year	Features	Method	Precision	Recall	Accuracy	TTM (sec)
Jain et al. [32]	2015	inside & outside(numerical)	AIO-HMM	77.4%	71.2%	N/A	3.53
Jain et al. [170]	2016	inside & outside(numerical)	RNN-LSTM	84.5%	77.1%	N/A	3.58
Moussaid et al. [171]	2019	inside & outside(numerical)	CNN-LSTM	N/A	N/A	94.1%	3.75
Tonutti et al. [173]	2019	inside & outside(numerical)	DA-RNN	92.3%	90.8%	N/A	3.98
Gite et al. [172]	2019	inside	RNN-LSTM	94.11%	97.56%	96.21%	4.89
Gebert et al. [174]	2019	inside & outside(video)	3D-ResNet	N/A	N/A	83.1%	0
Rong et al. [175]	2020	inside & outside(video)	ConvLSTM	N/A	N/A	83.98%	0
Zhou et al. [176]	2018	inside & outside(video)	LSTM	91.7%	90.7%	N/A	3.30

TABLE XXI

COMPARISON OF LANE CHANGE MANEUVER APPROACHES ON PRIVATE DATASETS. TIME-TO-MANEUVER (TTM), LARGER THE BETTER

Paper	Year	Camera View	Sensors	N Frames	Method	Accuracy	TTM(sec)
Doshi and Trivedi [142]	2008	1 face	CAN	NA	SVM	79.2	3
Martin et al. [177]	2018	2 face 1 hand	no	409K	CNN MVN	75	1
Xing et al. [178]	2020	1 face 1 hand 1 road	VBOX	270K	LSTM	96.1	0.5

dataset, which is commonly employed for lane change maneuver prediction.

Brain4Cars dataset [32] was collected from 10 different drivers, and contains 594 five-second video clips. Each maneuver data contains both inside features (driver face video) and outside features (outside video, number of empty lanes, speed information, GPS and street maps) with five types of maneuvers: left lane change, right lane change, left turn, right turn and straight drive. The properties of this dataset are summarized in Table XIX. Studies using the Brain4Cars dataset are summarized and compared in Table XX. The most commonly used metrics [32], namely precision, recall and time-to-maneuver (TTM) are presented in this table. Time-to-maneuver is defined as the time interval between the time of the model's prediction with the greatest confidence and the actual start of the maneuver.

Jain *et al.* [32] used a Hidden Markov Model to handle inside (2D facial features) and outside features at the same time for maneuver prediction. There are seven other papers using the Brain4Cars dataset, which are listed in Table XX and described below. They all rely on the LSTM-RNN model. Jain *et al.* [170] modified their method from a Hidden Markov Model to an LSTM-based model. They also updated the inside features from 2D facial features to 3D facial features. Moussaid *et al.* [171] and Gite *et al.* [172] extracted facial features from inside video and used LSTM to predict the final maneuver. Tonutti *et al.* [173] studied domain adaptation method on LSTM-GRU model to learn the shared components for leave-subjects-out lane change classification. Gebert *et al.* [174] and Rong *et al.* [175] used two branches, and processed inside and outside videos. One branch was used to encode RGB features, and another one used FlowNet to calculate optical flow. While Gebert *et al.* [174] computed optical flow from the inside video, Rong *et al.* [175] computed it from the outside video. These two papers are different from previous methods, which used numerical data, such as number of empty lanes, speed and road artifact, provided by other sensors as the outside features. These two papers, on the other hand, extracted the outside features from an outside camera. More specifically, they used CNN models to extract the outside features directly from the outside videos. They presented lower accuracy when time-to-maneuver is larger than 1 sec., compared to previous algorithms. Zhou *et al.* [176] used the same setup as the above two papers [174], [175], and also has a two-branch structure. They proposed a delay unit on the outside video branch, based on the observation that people have a delay in responding to the outside environment.

2) Studies Using Private Datasets: Studies in this category captured and used their own datasets to predict lane changes. These works are summarized in Table XXI together with some properties of each dataset. The common setup for lane change prediction usually involves a face-view camera. Doshi *et al.* [142] used Viola-Jones face detector to detect and track head motion from one camera facing the driver. They classified eye gaze into one of nine possible locations to detect where the driver was looking at. Then, they fed 1 sec. of this data into a relevance vector machine (the Bayesian counterpart to the Support Vector Machines (SVM)) to predict driver intention to change lanes. Martin *et al.* [177] fed data from two cameras (face view and hands view) into Alexnet to detect face, and then estimate the facial landmarks. Based on these data, features including head pose and horizontal gaze surrogate were extracted and used to estimate the driver gaze. They accumulated these data for 10 seconds, and analyzed them to obtain the frequency and duration of glance in each zone. They trained an unnormalized MVN, which used the Mahalanobis distance to estimate the driver intention. Xing *et al.* [178] collected data using multi-modal signals based on three cameras and a Velocity BOX (VBOX). Three cameras provided the driver's face view, hand view and road view. VBOX is a GPS receiver, which can record the car's dynamical data, including velocity, acceleration, heading, and positions of the vehicle. Collected data were labeled as lane-keeping, left lane change or right lane change maneuvers. They used the ensemble voting structure to take the outputs from multiple sub-BiLSTM modules, and used the majority label as the final maneuver. Facial landmarks and head positions were extracted from facial view video by Conditional Local Neural Fields (CLNF).

IV. CONCLUSION

This survey paper reflects a comprehensive effort to present and discuss the potential of naturalistic driving studies (NDS) and state-of-the-art computer vision techniques for driver

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behavior classification. The objective has been to establish a baseline accuracy in using deep learning models for driver behavior analysis, understand the impact of sensor placement on accuracy and establish the gaps in the current state-of-the-art techniques. This survey paper focused on works identifying driver behavior (low-level or high-level) with the use of at least one in-vehicle camera in real-car settings. Our findings are summarized in the following paragraphs arranged by low-level and high-level driver behavior.

Detecting a low-level behavior, such as head or eye movement, is the first step towards extracting information from NDS raw video data. Estimating head orientation and eye gaze helps to assess a driver's attention on the road. It is one way to classify a distracted driver versus an attentive driver. The LISA Gaze and DGW datasets are publicly available with ground-truth labels of significant regions of driver's gaze during driving. Across the reviewed papers, Convolutional Neural Networks (CNN) and Deep Neural Networks (DNN) have been found the most commonly implemented techniques to predict driver gaze, achieving a baseline accuracy of approximately 95% on the LISA Gaze dataset, 83% on the DGW dataset and above 90% on other private datasets. Among the reviewed studies, sensor placement is not observed to clearly impact the accuracy or performance of the model. Both single-camera-view and multiple-camera-view approaches have been found to perform well achieving accuracy values between 90-95%. The existent gap identified in gaze estimation studies is the lack of standard gaze zones within the vehicle. Even among the public datasets, there are only four similar classes of gaze zones. Future gaze estimation studies must focus on creating a comprehensive list of gaze zones covering the front, center and rear sides of the vehicle the driver will potentially sight during travel.

As for the head movement, three datasets, namely SHRP2, VIVA, and FDUD, are publicly available for estimating head pose and head orientation. Approaches ranging from conventional methods (the Viola-Jones algorithm) to CNN-based frameworks have been implemented for facial detection and head pose estimation. CNN-based models have reported better performance achieving a baseline accuracy of 88-99%, and have demonstrated the ability to overcome challenges due to lighting conditions, occlusions, and varying viewpoints. In terms of sensor placement, the three-camera framework was found to lower the error rate in estimating driver's head pose. However, for face detection, both single camera-view and three camera-view studies have achieved high accuracy. The existent gap in facial feature detection studies is developing techniques to detect driver's face overcoming challenges of occlusion due to weather or insufficient illumination and low resolution videos. The current review has shown that low-level driver behavior detection has been extensively studied. These low-level studies have been popularly conducted as a prior to extract features for high-level classification. Recent studies have focused on the detection of facial features in real-time, paving the way for driver assist systems to be used in autonomous vehicles.

An important rule in safe driving is keeping one's eyes on the road and hands on the wheel. Anything that disrupts a driver's attention to the road can result in accidents. Such contributing factors can be voluntary (e.g., texting, calling, eating) or involuntary in nature (e.g., fatigue, drowsiness). Drowsiness is a serious hindrance to driving caused by a driver undergoing physiological changes. In this survey paper, studies have been reviewed, which utilized visual features for drowsiness detection. Three public datasets, namely YawDD, DROZY and NTHU, have been used by implementing popular techniques including SVMs, HoG, CNNs, and LSTMs that follow a similar pipeline starting with the detection of facial features, such as eye or mouth, followed by the estimation of a measure (e.g., open or closed or duration of opening, blinking, or yawning) to classify the driver as attentive or drowsy. Across the different studies reviewed, deep learning techniques performed better than traditional methods, achieving a baseline accuracy of 85-96%. In terms of sensor placement, drowsiness studies are prominently carried out with single-camera-view datasets. Additionally, sensor data such as movement of the pedals and steering wheel, driver's electrocardiogram and respiration recordings and seat measure distribution are measured and used in drowsiness detection. The existent gap in literature is the need to explore techniques to study drowsiness using other facial features, since yawning detection is the most popular technique implemented to detect drowsiness.

Public datasets, such as the SHRP2, StateFarm DDD and AUC provide rich databases of events indicating a distracted driver. Studies using these datasets have aimed to classify the driver's behavior as distracted or attentive; in some cases, they have indicated the driver's specific activity. CNN-based models, which are used to detect distracted drivers, have been reported to provide high prediction accuracy of up to 99.68% with the State Farm dataset. The SHRP2 dataset has primarily been used for object detection to infer whether a driver is distracted or not. Proposed models using the SHRP2 dataset have detected features such as the driver's hand or a cell phone, with approximately 90% accuracy. Besides public datasets, many studies have used their private datasets to study distracted driving behavior and achieved a baseline accuracy of 80-97%. In terms of sensor placement, these datasets differ in the number of cameras, viewpoints, and additional sources of data available. Across these private datasets, different machine learning or deep learning techniques have helped achieve substantial accuracy above 90%. The existent gap in the literature is the lack of standardization for different labels identifying a distracted behavior. Learning from the public datasets available, future studies can aim to identify all the different activities listed as distracted driving.

Another essential element of driving behavior analysis vital to Advanced Driver Assistance Systems (ADAS) is lane change prediction. Brain4Cars is a commonly used public dataset to predict lane change intent, and detect lane change. The reviewed studies in general follow the framework of feature extraction and maneuver state prediction. Lane change prediction uses features from both inside and outside the vehicle, which is different from other driver behavior studies. CNN and LSTM are the two most popular techniques implemented, reporting a baseline accuracy of 83-90% for lane change prediction. In terms of sensor placement, studies analyzing

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more than the face view of the driver have also performed well predicting lane change maneuvers. The existent gap in the literature is the lack of public datasets besides the Brain4Cars dataset. More public datasets will lead the way for different techniques for feature extraction and lane change prediction providing a better comparison to identify the best technique for the objective.

Across the different high-level tasks discussed, recent studies are in favor of using deep learning models. Across the three high-level tasks, distracted driving has been extensively studied (51 papers). In contrast, there have been limited studies conducted on drowsiness (21 papers) and lane change detection (11 papers). Facial feature extraction is essential for studying driver behavior; however, there has been an imbalance in studying all parts of the face. Head pose and eye gaze studies are prevalent, while there has been limited research concerning the nose, mouth, or ears. We have observed that more complicated tasks have tended to be more DNN-based and that these models have had better performance than more basic models. CNN models have had greater popularity in drowsiness detection and distraction detection. However, for a more complicated task such as lane change prediction, CNN's implementation is not sufficient, and LSTM models are often additionally employed to investigate more features promptly.

It is evident that the pipeline to classifying a driver behavior is well-known to researchers, yet some of the techniques have fallen short in overcoming challenges in various datasets due to weather, illumination or a driver's physical characteristics. The biggest drawback with private dataset studies has been their lack of information about participants, driving times and trajectory. In the near future, it is important for researchers to develop an inclusive dataset accounting for these challenges in the real-world implementation of driver behavior classification models. It is also important to explore the possibilities of deploying models in real-time and building them with the capability to handle incoming video streams of a driver, and analyzing this video with minimum latency to relay the results. In the near future, we believe these areas will remain the prime focus of NDS research.

This survey paper has made a concerted effort to serve both NDS researchers and computer vision scientist communities. From this survey's review of the state-of-the-art techniques and analysis of available datasets, computer vision scientists stand to gain deep insight into what is currently known about how best to handle the modeling of driver behavior. On the other side, this review's analysis of data collection efforts and the designs of popular public datasets as well as private datasets is aimed at promoting discussion among NDS researchers about how best to streamline their efforts in future studies.

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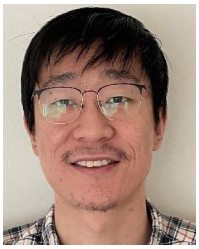
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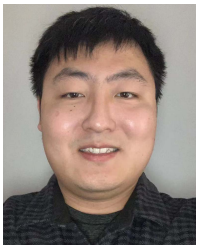
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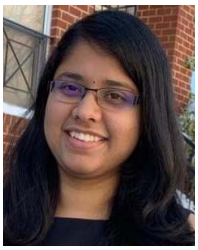
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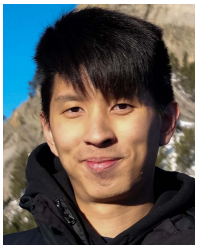
Jiyang Wang received the bachelor's degree in electrical engineering from the Anhui University of Science and Technology in 2017 and the master's degree in electrical engineering from Syracuse University in 2019, where he is currently pursuing the Ph.D. degree in electrical and computer engineering. His research interests include computer vision, deep learning, and human–computer interaction.



Weiheng Chai received the bachelor's degree in applied physics from the University of Science and Technology of China in 2016 and the master's degree in computer engineering from Syracuse University in 2019, where he is currently pursuing the Ph.D. degree in electrical and computer engineering. His research interests include computer vision, deep learning, and adversarial attacks.



Archana Venkatachalapathy received the bachelor's degree in civil engineering from the Visvesvaraya National Institute of Technology, India, in 2017, and the master's degree in civil engineering from Iowa State University in 2019, where she is currently pursuing the Ph.D. degree specializing in intelligent infrastructure engineering. Her research interests include naturalistic driving studies, driver stress behavior, and machine learning in transportation safety.



Kai Liang Tan received the bachelor's degree in mechanical engineering from Iowa State University in 2017, where he is currently pursuing the Ph.D. degree in mechanical engineering. He is a member of the Self-aware Complex Systems Lab, Iowa State University. His research interests include deep learning, deep reinforcement learning, computer vision to solve engineering problems, cyber-physical systems, and autonomous systems.



Arya Haghighat received the bachelor's degree in mechanical engineering from the Amirkabir University of Technology in 2014 and the master's degree from Iowa State University in 2019, where he is currently pursuing the Ph.D. degree in intelligent infrastructure engineering co-majored in mechanical engineering. His research interest includes the application of deep learning and computer vision in engineering problems and ITS.



Senem Velipasalar (Senior Member, IEEE) received the B.S. degree in electrical and electronic engineering from Bogazici University, Istanbul, Turkey, in 1999, the M.S. degree in electrical sciences and computer engineering from Brown University, Providence, RI, USA, in 2001, and the M.A. and Ph.D. degrees in electrical engineering from Princeton University, Princeton, NJ, USA, in 2004 and 2007, respectively. From 2007 to 2011, she was an Assistant Professor with the Department of Electrical Engineering, University of Nebraska-Lincoln. She is currently a Professor with the Department of Electrical Engineering and Computer Science, Syracuse University. Her research focus has been on machine learning, mobile camera applications, wireless embedded smart cameras, multicamera tracking, and surveillance systems.



Yaw Adu-Gyamfi received the master's and Ph.D. degrees in civil engineering from the University of Delaware. He is an Assistant Professor with the Department of Civil and Environmental Engineering, University of Missouri, Columbia. His research interests include sensor fusion, computer vision, machine learning, and big data analytics.



Anuj Sharma received the M.S. degree in Civil Engineering from Texas A&M University in 2004 and the Ph.D. degree in civil engineering from Purdue University in 2008. He was an Assistant Professor with the Department of Civil Engineering, University of Nebraska-Lincoln. He is currently a Pitts-Des Moines Inc. Professor in civil engineering at Iowa State University. His research focuses on traffic operations, big data analytics, machine learning, and traffic safety.

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